Introduction to parallel computing and OpenMP in C++ Part 2: shared variables

This activity is a pair programming exercise. This means that one person is at the keyboard (the driver) and the other is the navigator, observing, reviewing, and offering advice. This activity introduces the concept of a race condition and shared variables.

1. Starting with the code from part 1 (populating the array), that you did last week, add a new `for` loop that traverses the array, adding up all the values in the array using an accumulator. Check to make sure you are getting the correct answer using the formula for the sum of numbers from 1 to N is given by N(N+1)/2 (although the way we have populated the array, we are actually doing the sum from 0 + 1 + 2 + …. + N-1. Be sure that your code works for an N of 20,000,000. (This should be a non-parallel program with no openMP.)
2. Add the `#pragma omp parallel for` from Part1 to both loops in the program. Are you still getting the correct answer?
3. Change your array size N to a smaller value. Examine the accumulator variable inside the accumulation loop along with the thread id for each part of this loop. What is going on here?

TURN IN: Code that does this task in parallel and in serial. The serial code should work. Make sure you have implemented the formula to prove that it works and you get the correct value! Each block of code should also have timing information.